Problem Set 3 12209095 杜奕旋

Q1.

1. ① U has omitted variables which are correlated with the logyield and the rental_out_share. ∵ $E(\tilde{\beta}_1) = \beta_1 + \beta_2\tilde{\delta}_1$ ∴ If the omitted variable is positive correlate with rental_out_share, then direction is upwards; If is negative correlated, the direction is downwards

② rental_out_share; is measured with error: $\tilde{X}_i = X_i + e_i$. If $e_i > 0$, downwards; If $e_i < 0$, upwards

③ logyield causes rental_out_share and rental_out_share causes logyield

2. ① relevance: corr(total_land_area, rental_out_share) ≠ 0

② Exogeneity: corr(total_land_area, U) = 0

IV is true: the total land area is uncorrelated with any other factor, it is an objective variable. And has effect on rental_out_share.

3. $rental_out_share_i = \pi_0 + \pi_1(av_rain_i) + v_i$

$\widehat{rental_out_share_i} = \hat{\pi}_0 + \hat{\pi}_1(av_rain_i)$ $i = 1, 2, \ldots n$

```
. reg share100_rent_out av_rain
```

Source	SS	df	MS		
Model	174454.947	1	174454.947	Number of obs	= 13,897
Residual	66807981.2	13,895	4808.0591	F(1, 13895)	= 36.28
				Prob > F	= 0.0000
				R-squared	= 0.0026
				Adj R-squared	= 0.0025
Total	66982436.1	13,896	4820.26742	Root MSE	= 69.34

share100_r~t	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
av_rain	.0079141	.0013139	6.02	0.000	.0053388	.0104895
_cons	1.25257	1.345475	0.93	0.352	-1.384742	3.889883

```
. esttab, cells("b se t p") nogaps
```

	(1) share100_r~t			
	b	se	t	p
av_rain	.0079141	.0013139	6.023609	1.75e-09
_cons	1.25257	1.345475	.9309502	.3518955
N	13897			

∴ $\widehat{rental_out_share} = \underset{(1.345475)}{1.25257} + \underset{(.0013139)}{.0079141}(av_rain)$: first stage regression model

P-value for $\pi_1 = 1.75 \times 10^{-9}$ ∴ $H_0: \pi_1 = 0$ is rejected

∴ av_rain is correlated with rental_out_share

π_1 statistical significance = 1.75×10^{-7} %

π_0 statistical significance = 35.1896 %

Interpret: the rental_out_share change 0.0079141 when av_rain change 1 unit, while others stay same

4. Second stage regression model: $logyield_i = \beta_0 + \beta_1 \widehat{rental_out_share} + u_i$

```
. drop if missing( predict_rental_out_share ) | missing(av_rain) | missing(logyield) |missing( share100_rent_out )
(437 observations deleted)

. reg logyield predict_rental_out_share

      Source |       SS           df       MS      Number of obs   =    13,897
-------------+----------------------------------   F(1, 13895)     =      1.24
       Model |  .443552753         1  .443552753   Prob > F        =    0.2662
    Residual |  4985.20884    13,895  .358777174   R-squared       =    0.0001
-------------+----------------------------------   Adj R-squared   =    0.0000
       Total |  4985.65239    13,896  .358783275   Root MSE        =    .59898

------------------------------------------------------------------------------
    logyield | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
predict_re~e |   .0015945   .0014341     1.11   0.266    -.0012164    .0044055
       _cons |   5.924119   .0132614   446.72   0.000     5.898125    5.950113
------------------------------------------------------------------------------
```

```
. esttab, cells("b se t p") nogaps

------------------------------------------------------------
                      (1)
                 logyield
                        b            se             t            p
------------------------------------------------------------
predict_re~e       .0015945     .0014341     1.111886     .2662064
_cons              5.924119     .0132614     446.7189            0
------------------------------------------------------------
N                     13897
------------------------------------------------------------
```

$\therefore \widehat{logyield}_i = \underset{(.0132614)}{5.924119} + \underset{(.0014341)}{.0015945}\ \widehat{predicted\ rental_out_share}$

The logyield will change 0.0015945 when rental_out_share change 1 unit, while other stays the same.

β_0 p-value : ≈ 0

β_1 p-value : $.2662064 = 26.62\ \%$

5.

```
. ivreg logyield ( share100_rent_in = av_rain )

Instrumental variables 2SLS regression

      Source |       SS           df       MS      Number of obs   =    13,897
-------------+----------------------------------   F(1, 13895)     =      0.97
       Model |  -1339.55094         1 -1339.55094   Prob > F        =    0.3236
    Residual |   6325.20333    13,895  .455214345   R-squared       =         .
-------------+----------------------------------   Adj R-squared   =         .
       Total |   4985.65239    13,896  .358783275   Root MSE        =    .6747

------------------------------------------------------------------------------
    logyield | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
share100_re~n |   .0061052   .0061849     0.99   0.324    -.0060181    .0182284
       _cons |   5.910018   .0286602   206.21   0.000     5.85384     5.966196
------------------------------------------------------------------------------
Endogenous: share100_rent_in
Exogenous:  av_rain
```

$\widehat{logyield} = \underset{(.0286602)}{5.910018} + \underset{(5.910018)}{.0061052}\ rental_out_share$

```
. esttab, cells("b se t p") nogaps

------------------------------------------------------------
                      (1)
                 logyield
                        b            se             t            p
------------------------------------------------------------
share100_r~n       .0061052     .0061849     .9871083     .3236067
_cons              5.910018     .0286602     206.2099            0
------------------------------------------------------------
N                     13897
------------------------------------------------------------
```

The coefficients are likely the same between two regressions. The standard error in (4) is smaller than in (5) because standard error in (4) is not correct for it ignore the error in first stage.

6. (1) ① relevant assumption:

$rental_out_share_i = \pi_0 + \pi_1 Z_1 + \pi_2 Z_2 + V_i$

$H_0: \pi_1 = \pi_2 = 0 \quad H_1: \pi_1 \neq 0, \pi_2 \neq 0$

```
. reg  rental_out_share av_rain implemented

      Source |       SS           df       MS      Number of obs   =    13,862
-------------+----------------------------------   F(2, 13859)     =     25.08
       Model |  241493.659         2  120746.83     Prob > F        =    0.0000
    Residual |  66718577.1    13,859  4814.09749    R-squared       =    0.0036
-------------+----------------------------------   Adj R-squared   =    0.0035
       Total |  66960070.8    13,861  4830.82539    Root MSE        =    69.384

------------------------------------------------------------------------------
rental_out~e |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
    av_rain  |   .0075401   .0013201     5.71   0.000     .0049525    .0101276
implemented  |   4.774689   1.269582     3.76   0.000     2.286137    7.263242
       _cons |  -1.654896    1.55631    -1.06   0.288    -4.705475    1.395683
------------------------------------------------------------------------------
```

$\widehat{rental_out_share} = -1.654896 + .007540\,av_rain + 4.774689\,implemented$
$\quad\quad (1.55631) \quad\quad (.0013201) \quad\quad (1.269582)$

```
. test av_rain implemented

 ( 1)  av_rain = 0
 ( 2)  implemented = 0

       F(  2, 13859) =   25.08
            Prob > F =    0.0000
```

(2) $F = 25.08 > 10$

∴ The instrumental variables are not weak and H_0 is rejected. So is relevant

② Check exogenous assumption:

$Cov(Z_1, u) = 0$

$Cov(Z_2, u) = 0$

(1)

$\widehat{logyield_i} = \beta_0 + \beta_1 \widehat{rental_out_share} + u$

$\widehat{logyield} = 5.910018 + .006105\,\widehat{rental_out_share}$
$\quad\quad (.028602) \quad\quad (.0018449)$

step 1: Obtain the 2SLS estimator using all 2 instruments; compute the predicted value $\widehat{logyield}$ using actual rental_out_share's.

step 2: compute the residual $\hat{u}_i = logfield - \widehat{logfield}_i$

step 3: regress $\hat{u}_i$ on Z_{1i}, Z_{2i} $\quad \hat{u}_i = \delta_0 + \delta_1 av_rain + \delta_2 implemented$

step 4: compute the F-statistic testing the hypothesis that the coefficients on Z_{1i}, Z_{2i} are all zero $\quad H_0: \delta_1 = \delta_2 = 0 \quad H_1: \delta_1 \neq 0, \delta_2 \neq 0$

step 5: The J-statistic is $J = mF$.

(2)

step 1:

```
. ivreg logyield ( rental_out_share = av_rain implemented )

Instrumental variables 2SLS regression

      Source |       SS           df       MS      Number of obs   =    13,862
-------------+----------------------------------   F(1, 13860)     =     12.89
       Model | -1123.10207         1 -1123.10207    Prob > F        =    0.0003
    Residual |  5070.87301    13,860  .365863854    R-squared       =         .
-------------+----------------------------------   Adj R-squared   =         .
       Total |  3947.77094    13,861  .284811409    Root MSE        =    .60487

------------------------------------------------------------------------------
    logyield |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
rental_out~e |   .0044194   .0012309     3.59   0.000     .0020067    .006832
       _cons |   5.911039    .011707   504.92   0.000     5.888092    5.933986
------------------------------------------------------------------------------
Endogenous:  rental_out_share
Exogenous:   av_rain implemented
```

step 2:

```
. predict residuals, residuals
(2 missing values generated)
```

step 3:

$\hat{u} = -.00287 - .000141\,av_rain + .019465\,implemented$
$\quad\quad (.0135658) \quad\quad (.0000115) \quad\quad (.0110665)$

```
. reg residuals av_rain implemented

      Source |       SS           df       MS      Number of obs   =    13,862
-------------+----------------------------------   F(2, 13859)     =     2.15
       Model |  1.57609396         2 .788046982    Prob > F        =    0.1160
    Residual |  5069.2969    13,859 .365776528    R-squared       =    0.0003
-------------+----------------------------------   Adj R-squared   =    0.0002
       Total |  5070.873     13,861 .365837457    Root MSE        =    .60479

------------------------------------------------------------------------------
   residuals |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
    av_rain  |  -.0000141   .0000115    -1.23   0.220    -.0000367    8.43e-06
implemented  |   .0194657   .0110665     1.76   0.079    -.0022262    .0411576
       _cons |   -.000287   .0135658    -0.02   0.983    -.0268779    .0263038
------------------------------------------------------------------------------
```

step 4:

```
. test implemented av_rain

 ( 1)  implemented = 0
 ( 2)  av_rain = 0

       F(  2, 13859) =    2.15
            Prob > F =    0.1160
```

$J = 2F = 4.30$, p-value = 11.6% > 5%, so H_0 is not rejected.